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Roll No

EX-7101 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
Reliability Engineering
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. A random sample of 15 valves is observed. From past experience, it is known that the probability of a given failure within 500 hours following maintenance is 0.18. Calculate the probability that these valves will experience 0, 1, 2, ----, 15, independent failures within 500 hours following their maintenance.
2. Explain the state enumeration method for evaluating failure frequency.
3. What do you mean by random variable? Explain the distribution function of continuous and discrete random variable.
4. a) What are important time to failure life distributions? Discuss their characteristics and major applications in component reliability.
b) A stockpile of 40 relays contains 8 defective relays. If five relays are selected at random and the number of defective relays is known to be greater than two, what is the probability that exactly four relays are defective?
5. a) Explain the network reduction / solution techniques used in approximate system reliability evaluation.
b) Define minimal cut-set. Explain the reliability analysis of complex system using minimal cut-set method.
6. Derive the reliability function and MTTF for three CFR components in parallel. If three components, each with CFR are placed in parallel, determine the system reliability for 0.1 yr and the MTTF. Their failure rates are 5 per year, 10 per year and 15 per year.
7. A series system has 10 identical components. If the overall system reliability must be at least 0.99, what is the minimum reliability required of each component.

OR

Explain Markov analysis in detail. How reliability can be calculated from this model?

8. Write short notes on the followings (Any two)
 - a) Unavailability function
 - b) Parametric methods
 - c) Complex network
 - d) Replacement model

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